

A Godunov-type tensor artificial viscosity for staggered Lagrangian hydrodynamics

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ABSTRACT

This paper describes a tensor artificial viscosity for staggered Lagrangian hydrodynamics. Specifically, two viscous tensors are constructed using a discrete velocity gradient method. Under the condition of a piecewise constant distribution of velocity in each subcell, the Generalized Riemann Invariant relation, in the spirit of the Godunov methods, is applied to determine the coefficients of the viscous tensors. The artificial viscosity is found to be dissipative. Besides, the cylindrical symmetry of the developed artificial viscosity is demonstrated in an equi-angular polar grid. Typical numerical cases with strong shocks show that the tensor artificial viscosity is robust and performs well in different grid types.

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1. Introduction

Lagrangian hydrodynamics, in which the computational grid moves simultaneously with the fluid, is widely used for multi-material flow simulations. Its property of resolving sharp contact discontinuities has a broad range of applications in fields such as astrophysics and inertial confinement fusion.

Early investigations [1] led to the development of staggered grid hydrodynamics (SGH), which employs a staggered placement of variables. In SGH, the momentum equation is solved around a nodal control volume, while the energy equation is solved around a zonal control volume. A classical SGH scheme [2] uses the support operator method [3], which discretizes the governing equations in a compatible way. The advantage of this scheme is that it exactly ensures the conservation of momentum and total energy. In the presence of shock waves in the SGH, an artificial viscosity is needed to eliminate the possible numerical oscillations. For 1D calculations, Von Neumann and Richtmyer [1] introduced a viscosity with a quadratic term that is added to the pressure in the momentum and energy equations. Later, Landshoff [4] proposed an additional linear term that improved Von Neumann's method by ensuring fewer oscillations behind the shock. Another well-known artificial viscosity proposed by Kurapatenko [5] uniformly matches the quadratic term and the linear term. This artificial viscosity is a good complement of the formulation proposed by Landshoff. For 2D and 3D calculations, many artificial viscosities have been established on the basis of the 1D artificial viscosity with minor modifications. Advanced multi-dimensional artificial viscosities can be edge based [6], tensorial [7–9], or Laplacian-based [10].

A different approach designed by Godunov [11] employs a collocated placement of all variables in one control volume. This Godunov-type method is known as cell-centered hydrodynamics (CCH) in the Lagrangian framework. Classically, such schemes overcome the discontinuities at cell edges using an approximate Riemann solver. A remarkable study using CCH was conducted by Despres [12], who constructed a node solver based on a Riemann invariant relation that satisfies the

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time-averaged equations for momentum conservation, total energy conservation, geometric conservation law (GCL), and entropy inequality. Subsequently, Maire [13,14] established a sophisticated node solver that exactly recovers the 1D acoustic solver by allocating two pressures on each subcell. In [15], Burton considers the ancillary equation of rotational equilibrium to establish a new node solver with increased mesh robustness. For more complex discontinuous conditions, Jia [16] improved Maire's work by recovering the numerical flux in the Harten–Lax–van Leer contact (HLLC) scheme using the finite element framework. Based on the above node solvers, important progress in CCH has been made in many areas (e.g., 2D/3D numerical schemes, planar/cylindrical coordinates, finite volume/element frameworks) [17–22].

According to previous investigations, the artificial viscosity and approximate Riemann solution are two different shock-capturing mechanisms. However, their gaps have been bridged by the introduction of Godunov-type artificial viscosities. Related works can be traced back to the application of Godunov's artificial viscosity scheme in the early 1990s. Christensen [23] noticed that the artificial viscosity in SGH can be obtained by the HLL approximate Riemann solver under certain conditions. Luttwak et al. [24,25] proposed to use a uniaxial tensor pseudo-viscosity based on Riemann problem in staggered meshes. Their method is known as the Staggered Mesh Godunov scheme. Later, a compatible Lagrangian discretization and associated Riemann solver-based artificial viscosity were developed in both the 2D [26–28] and 3D [29] cases. As such artificial viscosity specifies the jump condition explicitly, high-order accuracy in space can be extended through Taylor expansions of the velocity field. Detailed information can be found in the implementation of a second-order least-squares approach with a frame-invariant limiter [14]. A further potential benefit of such viscosity is that, as long as the connection between the viscosity and the Riemann solution is determined, different types of Riemann solvers (e.g., HLL, HLLC, HLL-E) can capture the shock in high resolution. However, a disadvantage of the Godunov-type artificial viscosity is that little attention has been paid to the nature of its tensor. As a result, inaccuracies arise in the viscosity discretization, and these might amplify the hourglass instabilities with spurious grid distortion [8].

In this paper, the general form of the artificial viscosity is derived in the full tensor space using a discrete gradient operator method. Moreover, the relationship between the artificial viscosity and the Generalized Riemann Invariant relation is presented, allowing us to determine the viscosity coefficients. Detailed analysis shows that the developed tensor artificial viscosity is superior to the Godunov-type scalar viscosity [27] in that it is applicable for an arbitrary distribution of the velocity gradient. A series of numerical tests containing strong discontinuities demonstrates the robustness of the developed viscosity.

The remainder of this paper is organized as follows. Section 2 briefly introduces the SGH discretizations that satisfy the total energy conservation condition. In Section 3, the theory and derivation of the Godunov-type artificial viscosity are described in detail. The symmetry of the developed viscosity is proved in Section 4. In Section 5, typical numerical examples are used to verify the performance of the Godunov-type artificial viscosity. Finally, Section 6 presents the conclusions to this study.

2. Governing equations and discretizations

2.1. Governing equations

In a 2D Lagrangian frame, the gas dynamics equations are

$$\rho \frac{d}{dt} \left(\frac{1}{\rho} \right) - \nabla \cdot \mathbf{u} = 0, \quad (1a)$$

$$\rho \frac{d\mathbf{u}}{dt} + \nabla p = 0, \quad (1b)$$

$$\rho \frac{dE}{dt} + \nabla \cdot (p\mathbf{u}) = 0, \quad (1c)$$

where ρ is the density, p is the pressure, $\mathbf{u}$ is the fluid velocity, and E is the total energy. Eq. (1a) refers to the GCL, which requires the rate of change of a Lagrangian volume to be consistent with the node motion. Eqs. (1b) and (1c) are the momentum and total energy conservation equations, respectively.

For a smooth flow, the total energy conservation equation can be replaced by the internal energy equation

$$\rho \frac{d\varepsilon}{dt} + p \nabla \cdot \mathbf{u} = 0, \quad (2)$$

where $\varepsilon = E - |\mathbf{u}|^2/2$ is the specific internal energy. Combined with Eq. (1a), this gives

$$\rho \frac{d\varepsilon}{dt} + p \rho \frac{d}{dt} \left(\frac{1}{\rho} \right) = 0. \quad (3)$$

For non-smooth flows, the second law of thermodynamics implies that $T \frac{dS}{dt} \geq 0$ for temperature T and specific entropy S . This entropy inequality can be rewritten using the Gibbs relation as

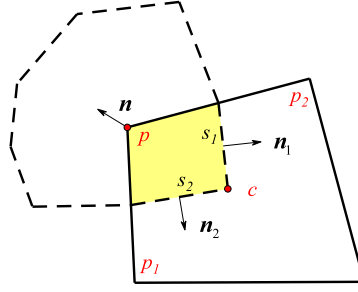


Fig. 1. Sketch and notation of a discrete unit in SGH.

$$\rho \frac{d\varepsilon}{dt} + p\rho \frac{d}{dt} \left(\frac{1}{\rho} \right) \geq 0. \quad (4)$$

Combining Eqs. (1a), (1b), and (2), the non-conservative system that is usually applied in the SGH is constructed. In the Lagrangian description of motion, the trajectory equations are written as

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}(t), t) \quad \mathbf{x}(0) = \mathbf{x}_0. \quad (5)$$

Adding an equation of state (EOS) completes the system. We consider the EOS of an ideal gas here, i.e.,

$$p = (\gamma - 1)\rho\varepsilon. \quad (6)$$

2.2. Notation and assumptions

As shown in Fig. 1, the kinematic variables are located at the cell vertices and the thermodynamic variables are located at the cell centers. Each quadrilateral cell is assigned an index c . The cell Ω_c is surrounded by a set of vertices labeled by $P(c)$. For a specific vertex p , its neighbors on the cell Ω_c are p_1 and p_2 . By connecting the centroid of Ω_c and p to the midpoints of the edges $[p, p_1]$, $[p, p_2]$ in sequence, we form the subcell Ω_{pc} . The union of Ω_{pc} related to the same vertex p constitutes the dual grid Ω_p , and $C(p)$ represents a set of cells that share the same vertex p .

Next, the mass of the subcell Ω_{pc} is set to be the primitive quantity to help us construct the total mass of the cell Ω_c and the dual grid Ω_p . Assuming the subcell Ω_{pc} is a Lagrangian volume, we have

$$m_c = \sum_{p \in P(c)} m_{cp} \quad m_p = \sum_{c \in C(p)} m_{cp}, \quad (7)$$

where m_{cp} , m_c , and m_p are the Lagrangian masses of the subcell Ω_{pc} , the cell Ω_c , and the dual grid Ω_p , respectively. The volume V_{pc} of the subcell Ω_{pc} is such that

$$V_c = \sum_{p \in P(c)} V_{cp} \quad V_p = \sum_{c \in C(p)} V_{cp}, \quad (8)$$

where V_c and V_p are the volumes of the cell Ω_c and the dual grid Ω_p , respectively. Let c^+ and c^- denote the midpoints of the edges $[p, p_1]$ and $[p, p_2]$, l_1 and l_2 be the lengths of the edges $[c, c^+]$ and $[c, c^-]$, and $\mathbf{n}_1$ and $\mathbf{n}_2$ be the unit outward normal vectors of the edges $[c, c^+]$ and $[c, c^-]$. A subcell vector is thus introduced as

$$\mathbf{s}_{cp} = l_1 \mathbf{n}_1 + l_2 \mathbf{n}_2. \quad (9)$$

2.3. Spatial discretization

The cornerstones of the spatial discretization are the discrete support operators. First, a discrete divergence operator is defined such that the discretization of the GCL equation (1a) is equivalent to the trajectory equations (6). Next, a discrete gradient operator is derived to be compatible with the discrete divergence operator. The subcell force is thus defined so as to discretize the momentum equation. Finally, the discretized internal energy equation is obtained using the total energy conservation.

2.3.1. Geometric conservation law

Integrating the GCL equation (1a) over the cell Ω_c , we obtain

$$\frac{1}{V_c} \frac{dV_c}{dt} = (\nabla \cdot \mathbf{u})_c, \quad (10)$$

where $(\nabla \cdot \mathbf{u})_c$ is the discrete divergence operator over the cell Ω_c . This operator can be expressed in face discretization form as

$$(\nabla \cdot \mathbf{u})_c = \frac{1}{V_c} \sum_{f \in F(c)} l_f^c \mathbf{n}_f^c \mathbf{u}_f^c, \quad (11)$$

where the index f denotes a generic face of cell Ω_c and $F(c)$ is the set of all faces of the cell. Further, l_f^c , $\mathbf{n}_f^c$, and $\mathbf{u}_f^c$ denote the length, unit outward normal, and velocity of each face. The rate of change of V_c can be expressed as a function of the vertex coordinates that constitute the cell Ω_c , that is,

$$\frac{dV_c}{dt} = \sum_{p \in P(c)} \frac{1}{2} (l_{pp_1} \mathbf{n}_{pp_1} + l_{pp_2} \mathbf{n}_{pp_2}) \cdot \mathbf{u}_p, \quad (12)$$

where l_{pp_1} and l_{pp_2} are the lengths of edges $[p, p_1]$ and $[p, p_2]$, and $\mathbf{n}_{pp_1}$ and $\mathbf{n}_{pp_2}$ are the unit outward normal vectors of these edges. By shifting the indices in the previous summation, Eq. (12) becomes

$$\frac{dV_c}{dt} = \sum_{f \in F(c)} l_f^c \mathbf{n}_f^c \cdot \frac{1}{2} (\mathbf{u}_{f_1}^c + \mathbf{u}_{f_2}^c), \quad (13)$$

where $\mathbf{u}_{f_1}^c$ and $\mathbf{u}_{f_2}^c$ are the velocities of the start point and end point of each face. A comparison of Eqs. (11) and (13) shows that they are equivalent under the condition that

$$\mathbf{u}_f^c = \frac{1}{2} (\mathbf{u}_{f_1}^c + \mathbf{u}_{f_2}^c). \quad (14)$$

With the sufficient condition expressed in Eq. (14), a discrete divergence operator that makes the GCL equation (1a) equivalent to the trajectory equations (6) is defined as

$$(\nabla \cdot \mathbf{u})_c = \frac{1}{V_c} \sum_{p \in P(c)} \frac{1}{2} (l_{pp_1} \mathbf{n}_{pp_1} + l_{pp_2} \mathbf{n}_{pp_2}) \cdot \mathbf{u}_p. \quad (15)$$

2.3.2. Momentum equation

The discrete divergence operator over the grid Ω_p is

$$(\nabla \cdot \mathbf{u})_p = \frac{1}{V_p} \sum_{f \in F(p)} l_f^p \mathbf{n}_f^p \mathbf{u}_f^p, \quad (16)$$

where the index f denotes a generic face of the grid Ω_p , and $F(c)$ is the set of faces of the grid. Further, l_f^p , $\mathbf{n}_f^p$, and $\mathbf{u}_f^p$ denote the length, unit outward normal, and velocity of each face. To ensure consistency with the GCL equation (1a) over the grid Ω_p , there exists a condition similar to Eq. (14), stated as

$$\mathbf{u}_f^p = \frac{1}{2} (\mathbf{u}_{f_1}^p + \mathbf{u}_{f_2}^p), \quad (17)$$

where $\mathbf{u}_{f_1}^p$ and $\mathbf{u}_{f_2}^p$ are the velocities of the start point and end point of each face.

Let us introduce two pressures related to the edges $[c, c^+]$ and $[c, c^-]$ in Fig. 1. These pressures are denoted as π_1 and π_2 , respectively. Thus, a discrete gradient operator $(\nabla p)_p$ can be defined in a manner that is compatible with the discrete divergence operator $(\nabla \cdot \mathbf{u})_p$ in Eq. (16), that is,

$$(\nabla p)_p = \frac{1}{V_p} \sum_{f \in F(p)} l_f^p \mathbf{n}_f^p p_f^p = \frac{1}{V_p} \sum_{c \in C(p)} (l_1 \pi_1 \mathbf{n}_1 + l_2 \pi_2 \mathbf{n}_2), \quad (18)$$

where p_f^p is the pressure of a generic face. Consequently, the semi-discrete momentum equation is obtained by integrating Eq. (1b) over the grid Ω_p :

$$m_p \frac{d}{dt} \mathbf{u}_p + \sum_{c \in C(p)} (l_1 \pi_1 \mathbf{n}_1 + l_2 \pi_2 \mathbf{n}_2) = 0. \quad (19)$$

At this point, we define the subcell force $\mathbf{f}_{cp}$, which acts on the grid Ω_p from the cell Ω_c , as

$$\mathbf{f}_{cp} = l_1 \pi_1 \mathbf{n}_1 + l_2 \pi_2 \mathbf{n}_2, \quad (20)$$

and the discretized momentum equation becomes

$$m_p \frac{d}{dt} \mathbf{u}_p + \sum_{c \in C(p)} \mathbf{f}_{cp} = 0. \quad (21)$$

2.3.3. Specific internal energy equation

For total energy conservation without considering the boundary conditions, the sum of the total kinetic energy and total internal energy is a constant function of time,

$$\frac{dE}{dt} = \frac{d}{dt} \sum_p \left(\frac{1}{2} m_p \mathbf{u}_p^2 \right) + \frac{d}{dt} \sum_c (m_c \varepsilon_c) = 0. \quad (22)$$

Combining Eqs. (21) and (22), we obtain the internal energy equation

$$m_c \frac{d}{dt} \varepsilon_c - \sum_{p \in P(c)} \mathbf{f}_{cp} \cdot \mathbf{u}_p = 0. \quad (23)$$

To summarize, the semi-discrete finite volume scheme with the primary variables $(\mathbf{u}_p, \varepsilon_c, \mathbf{x})$ is

$$m_p \frac{d}{dt} \mathbf{u}_p + \sum_{c \in C(p)} \mathbf{f}_{cp} = 0, \quad (24a)$$

$$m_c \frac{d}{dt} \varepsilon_c - \sum_{p \in P(c)} \mathbf{f}_{cp} \cdot \mathbf{u}_p = 0 \quad (24b)$$

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}(t), t) \quad \mathbf{x}(0) = \mathbf{x}_0. \quad (24c)$$

Note that the GCL equation is omitted because it is equivalent to the trajectory equations.

2.3.4. Galilean invariance and entropy inequality

According to the concept of Galilean invariance [27], in another inertial frame with a uniform translation velocity of $\mathbf{u}_c$, the specific internal energy equation (23) has the form

$$m_c \frac{d}{dt} \varepsilon_c - \sum_{p \in P(c)} \mathbf{f}_{cp} \cdot (\mathbf{u}_p - \mathbf{u}_c) = 0. \quad (25)$$

The above equation requires

$$\sum_{p \in P(c)} \mathbf{f}_{cp} = 0. \quad (26)$$

Using the Gibbs relation for the cell Ω_c , we have

$$m_c T_c \frac{d}{dt} S_c = m_c \left[\frac{d}{dt} \varepsilon_c + p_c \frac{d}{dt} \left(\frac{1}{\rho_c} \right) \right]. \quad (27)$$

Substituting Eqs. (24a) and (25) into Eq. (27) yields

$$\begin{aligned} m_c T_c \frac{d}{dt} S_c &= \sum_{p \in P(c)} \mathbf{f}_{cp} \cdot (\mathbf{u}_p - \mathbf{u}_c) - p_c \sum_{p \in P(c)} \mathbf{s}_{cp} \cdot \mathbf{u}_p \\ &= \sum_{p \in P(c)} \mathbf{f}_{cp} \cdot (\mathbf{u}_p - \mathbf{u}_c) - p_c \sum_{p \in P(c)} \mathbf{s}_{cp} \cdot (\mathbf{u}_p - \mathbf{u}_c) \\ &= \sum_{p \in P(c)} (\mathbf{f}_{cp} - \mathbf{s}_{cp} p_c) \cdot (\mathbf{u}_p - \mathbf{u}_c). \end{aligned} \quad (28)$$

To satisfy the local entropy condition, i.e., $m_c T_c \frac{d}{dt} S_c \geq 0$, a sufficient condition is to construct the subcell force in the following form [27]:

$$\mathbf{f}_{cp} = p_c \mathbf{s}_{cp} - M_{cp} \Delta \mathbf{u}_{cp}, \quad (29)$$

where M_{cp} is a subcell-based 2×2 non-negative semidefinite matrix, and $\Delta \mathbf{u}_{cp}$ is the difference between $\mathbf{u}_c$ and $\mathbf{u}_p$. The right-hand side of the above equation can be split into an isentropic part and a viscous part. In Section 3, the particular form of the viscous part is introduced through a Godunov-type artificial viscosity.

2.4. Time discretization

The momentum and internal energy equation are both integrated explicitly in time to ensure the exact conservation of total energy. The predictor–corrector method [30] is used to ensure the second-order time accuracy. With the given time step $\Delta t = t^{n+1} - t^n$, the quantities at time t^n are first used to update the governing equations in the predictor stage, that is,

$$m_p \frac{\mathbf{u}_p^{pr} - \mathbf{u}_p^n}{\Delta t} + \sum_{c \in C(p)} \mathbf{f}_{cp}^n = 0, \quad (30a)$$

$$m_c \frac{\varepsilon_c^{pr} - \varepsilon_c^n}{\Delta t} - \sum_{p \in P(c)} \mathbf{f}_{cp}^n \cdot \mathbf{u}_p^n = 0, \quad (30b)$$

$$\frac{\mathbf{x}_p^{pr} - \mathbf{x}_p^n}{\Delta t} = \frac{1}{2} (\mathbf{u}_p^{pr} + \mathbf{u}_p^n) = \mathbf{u}_p^{n+\frac{1}{2}}, \quad (30c)$$

where the superscript “pr” represents the predicted value.

After the predictor stage, the essentially half-time centered variables can be evaluated as

$$\mathbf{x}_p^{n+\frac{1}{2}} = \frac{1}{2} (\mathbf{x}_p^n + \mathbf{x}_p^{pr}), \quad (31a)$$

$$\rho_c^{n+\frac{1}{2}} = m_c / V(\mathbf{x}_p^{n+\frac{1}{2}}; p \in P(c)), \quad (31b)$$

$$p^{n+\frac{1}{2}} = \text{EOS} \left(\rho_c^{n+\frac{1}{2}}, (\varepsilon_c^{pr} + \varepsilon_c^n) / 2 \right), \quad (31c)$$

$$\mathbf{f}_{cp}^{n+\frac{1}{2}} = f \left(\mathbf{x}_p^{n+\frac{1}{2}}, \rho_c^{n+\frac{1}{2}}, p^{n+\frac{1}{2}}, \mathbf{u}_p^{n+\frac{1}{2}} \right). \quad (31d)$$

The half-time centered variables are then used to perform the corrector stage:

$$m_p \frac{\mathbf{u}_p^{n+1} - \mathbf{u}_p^n}{\Delta t} + \sum_{c \in C(p)} \mathbf{f}_{cp}^{n+\frac{1}{2}} = 0, \quad (32a)$$

$$m_c \frac{\varepsilon_c^{n+1} - \varepsilon_c^n}{\Delta t} - \sum_{p \in P(c)} \mathbf{f}_{cp}^{n+\frac{1}{2}} \cdot \mathbf{u}_p^{n+\frac{1}{2}} = 0, \quad (32b)$$

$$\frac{\mathbf{x}_p^{n+1} - \mathbf{x}_p^n}{\Delta t} = \frac{1}{2} (\mathbf{u}_p^{n+1} + \mathbf{u}_p^n). \quad (32c)$$

To close the n -th time loop, the initialized values of the thermodynamic variables for the $(n+1)$ -th time loop are calculated as

$$\rho_c^{n+1} = m_c / V(\mathbf{x}_p^{n+1}; p \in P(c)), \quad (33a)$$

$$p^{n+1} = \text{EOS}(\rho_c^{n+1}, \varepsilon_c^{n+1}), \quad (33b)$$

$$\mathbf{f}_{cp}^{n+1} = f(\mathbf{x}_p^{n+1}, \rho_c^{n+1}, p^{n+1}, \mathbf{u}_p^{n+1}). \quad (33c)$$

The time step is determined by the CFL condition and on a criterion based on the variation of volume. The details were described by Barlow [30].

3. Godunov-type artificial viscosity

Closely following Maire et al. [27], one can apply the Generalized Riemann Invariant relation in each subcell. This allows us to define an edge-based Godunov-type artificial viscosity. Through the surface integration of the artificial viscosity, a subcell-based matrix M_{cp} is specified. In addition, the above process involves a new degree of freedom due to the unknown cell center velocity $\mathbf{u}_c$. Using the Galilean invariance, $\mathbf{u}_c$ can then be determined iteratively. In this section, we will mostly follow this strategy. Somewhat different from Maire et al., the Generalized Riemann Invariant relation is established in the tensor space. Afterward, a Godunov-type tensor artificial viscosity and the resulting subcell-based matrix M_{cp} are presented. Specifically, Section 3.1 introduces the motivation for our work, Section 3.2 summarizes the first-order Godunov-type scalar artificial viscosity from Maire et al., Section 3.3 presents a full derivation of the first-order Godunov-type tensor artificial viscosity, and Section 3.4 describes the general reconstruction technique.

3.1. Motivation

Previously, in Eq. (29), the subcell force was split into an isentropic part and a viscous part:

$$\mathbf{f}^{ise} = p_c \mathbf{S}_{cp}, \quad (34a)$$

$$\mathbf{f}^{vis} = -M_{cp} \Delta \mathbf{u}_{cp}, \quad (34b)$$

where $\mathbf{f}^{ise}$ and $\mathbf{f}^{vis}$ represent the pressure force and the artificial viscous force, respectively. The artificial viscous force is usually approximated as the surface integration of the artificial viscosity, i.e.,

$$\mathbf{f}^{vis} = \int_{\partial \Omega_{cp}} \mathbf{Q} \cdot \mathbf{n} dS, \quad (35)$$

where $\mathbf{Q}$ represents the artificial viscosity. A comparison of Eqs. (34b) and (35) indicates that

$$\int_{\partial \Omega_{cp}} \mathbf{Q} \cdot \mathbf{n} dS = -M_{cp} \Delta \mathbf{u}_{cp}, \quad (36)$$

and it can be deduced that the artificial viscosity $\mathbf{Q}$ might have some tensorial features because M_{cp} is a 2×2 matrix. However, Maire et al. only considered the Godunov-type artificial viscosity as a scalar, which makes their approach simple, yet deserves more investigation. From another point of view, the tensor artificial viscosity is not a new idea. Research [7,8] shows that a tensor form of the artificial viscosity helps to reduce the grid dependence of the numerical results. Therefore, it is interesting to formulate the Godunov-type artificial viscosity as a tensor, and to examine whether this could improve the numerical results compared to the scalar form.

3.2. Godunov-type scalar artificial viscosity

In a previous study [27], the pressures π_1 and π_2 are regarded as scalars. To ensure consistency with the definition of the subcell force in Eq. (20), an edge-based artificial viscosity with two components is defined as

$$q_1 = \pi_1 - p_c, \quad (37a)$$

$$q_2 = \pi_2 - p_c, \quad (37b)$$

where q_1 and q_2 are the scalar viscosities related to edges $[c, c^+]$ and $[c, c^-]$, respectively. Using the Generalized Riemann Invariant relation [13], we obtain

$$p_c - \pi_1 = \alpha_1 \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1, \quad (38a)$$

$$p_c - \pi_2 = \alpha_2 \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2, \quad (38b)$$

where α_1 and α_2 are defined as Dukowicz's mass fluxes [31] across $[c, c^+]$ and $[c, c^-]$,

$$\alpha_1 = \rho_c [a_c + \Gamma_c |\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1|], \quad (39a)$$

$$\alpha_2 = \rho_c [a_c + \Gamma_c |\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2|], \quad (39b)$$

where a_c is the speed of sound in a cell. Γ_c is 0 for expansion and $(\gamma + 1)/2$ for compression. Thus, the values of q_1 and q_2 are

$$q_1 = -\alpha_1 \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1, \quad (40a)$$

$$q_2 = -\alpha_2 \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2. \quad (40b)$$

The surface integration of these viscosities exactly recovers the general form of M_{cp} in Eq. (36):

$$M_{cp} = \alpha_1 l_1 \mathbf{n}_1 \otimes \mathbf{n}_1 + \alpha_2 l_2 \mathbf{n}_2 \otimes \mathbf{n}_2. \quad (41)$$

However, $\mathbf{u}_c$ remains unknown in the expressions for q_1 , q_2 , and M_{cp} . Using the Galilean invariance in Eq. (26), we obtain

$$\sum_{p \in P(c)} M_{cp} \Delta \mathbf{u}_{cp} = 0. \quad (42)$$

Table 1
Values of M_{cp} and $\mathbf{f}_{cp}$ in [27] and our approach when $|\Delta \mathbf{u}_{cp}| \neq 0$.

	M_{cp}	$\mathbf{f}_{cp}$
Ref. [27]	$\alpha_1 l_1 (\mathbf{n}_1 \otimes \mathbf{n}_1) + \alpha_2 l_2 (\mathbf{n}_2 \otimes \mathbf{n}_2)$	$p_c \mathbf{s}_{cp} - (\alpha_1 l_1 \mathbf{n}_1 \otimes \mathbf{n}_1 + \alpha_2 l_2 \mathbf{n}_2 \otimes \mathbf{n}_2) \Delta \mathbf{u}_{cp}$
Our approach	$\left(\alpha_1 l_1 \frac{ \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1 }{ \Delta \mathbf{u}_{cp} } + \alpha_2 l_2 \frac{ \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2 }{ \Delta \mathbf{u}_{cp} } \right) I$	$p_c \mathbf{s}_{cp} - \left(\alpha_1 l_1 \frac{ \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1 }{ \Delta \mathbf{u}_{cp} } + \alpha_2 l_2 \frac{ \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2 }{ \Delta \mathbf{u}_{cp} } \right) \Delta \mathbf{u}_{cp}$

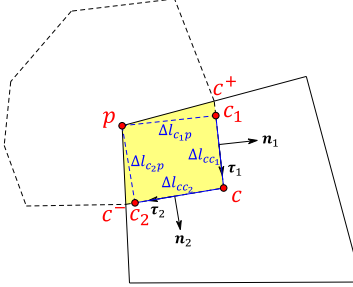


Fig. 2. Projections of the discrete velocity gradient in a subcell. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Thus, an iterative procedure is employed to compute $\mathbf{u}_c$, i.e.,

$$\mathbf{u}_c = M_c^{-1} \sum_p M_{cp} \mathbf{u}_p, \quad (43)$$

where M_c is defined as $M_c = \sum_{p \in P(c)} M_{cp}$.

Substituting Eq. (41) into Eq. (29), we have an expression for the subcell force, and this will be used to update the governing equations explicitly:

$$\mathbf{f}_{cp} = p_c \mathbf{s}_{cp} - (\alpha_1 l_1 \mathbf{n}_1 \otimes \mathbf{n}_1 + \alpha_2 l_2 \mathbf{n}_2 \otimes \mathbf{n}_2) \Delta \mathbf{u}_{cp}. \quad (44)$$

3.3. Godunov-type tensor artificial viscosity

This subsection shows that a Godunov-type tensor artificial viscosity can be defined in a similar way to the scalar form. However, a new expression of M_{cp} is derived to determine the subcell force $\mathbf{f}_{cp}$. Expressions for M_{cp} and $\mathbf{f}_{cp}$ are presented in Table 1. We now describe the approach used to derive these expressions in detail. Note that when $|\Delta \mathbf{u}_{cp}| = 0$, M_{cp} and the viscous part of $\mathbf{f}_{cp}$ are set to zero in our approach.

To derive a tensor artificial viscosity, let us first replace the scalar pressures π_1 and π_2 in Eq. (37) by the tensor pressures Π_1 and Π_2 . The tensor viscosities Q_1 and Q_2 related to the edges $[c, c^+]$ and $[c, c^-]$ can then be defined as

$$Q_1 = \Pi_1 - p_c I, \quad (45a)$$

$$Q_2 = \Pi_2 - p_c I. \quad (45b)$$

It is common practice to define the above tensor viscosities as a combination of a scalar coefficient and the velocity gradient [7], i.e.,

$$Q_1 = \mu_1 G_1, \quad Q_2 = \mu_2 G_2, \quad (46)$$

where μ_1 and μ_2 are edge-related scalar coefficients and G_1 and G_2 are edge-related velocity gradients. Fig. 2 depicts a subcell colored yellow. For the edges $[c, c^+]$ and $[c, c^-]$, the unit normal vectors are $\mathbf{n}_1$ and $\mathbf{n}_2$ and the unit tangential vectors are $\boldsymbol{\tau}_1$ and $\boldsymbol{\tau}_2$, respectively. Consider the perpendicular lines from point p to the edges $[c, c^+]$ and $[c, c^-]$, where the intersection points are c_1 and c_2 . The velocities of the intersection points are $\mathbf{u}_{c_1}$ and $\mathbf{u}_{c_2}$, respectively. The velocity differences are represented as $\Delta \mathbf{u}_{c_1p} = \mathbf{u}_{c_1} - \mathbf{u}_p$, $\Delta \mathbf{u}_{cc_1} = \mathbf{u}_c - \mathbf{u}_{c_1}$, $\Delta \mathbf{u}_{c_2p} = \mathbf{u}_{c_2} - \mathbf{u}_p$, and $\Delta \mathbf{u}_{cc_2} = \mathbf{u}_c - \mathbf{u}_{c_2}$. Let Δl_{c_1p} , Δl_{cc_1} , Δl_{c_2p} , and Δl_{cc_2} denote the lengths of the segments $[c_1, p]$, $[c, c_1]$, $[c_2, p]$, and $[c, c_2]$, respectively. Using the discrete velocity gradient method [7], the velocity gradients are represented as

$$G_1 = \frac{1}{\Delta l_{c_1p}} \Delta \mathbf{u}_{c_1p} \otimes \mathbf{n}_1 + \frac{1}{\Delta l_{cc_1}} \Delta \mathbf{u}_{cc_1} \otimes \boldsymbol{\tau}_1, \quad (47a)$$

$$G_2 = \frac{1}{\Delta l_{c_2p}} \Delta \mathbf{u}_{c_2p} \otimes \mathbf{n}_2 + \frac{1}{\Delta l_{cc_2}} \Delta \mathbf{u}_{cc_2} \otimes \boldsymbol{\tau}_2; \quad (47b)$$

see Appendix A for the derivation. Consequently, the tensor viscosities are represented as

$$Q_1 = \frac{\mu_1}{\Delta l_{c_1 p}} \Delta \mathbf{u}_{c_1 p} \otimes \mathbf{n}_1 + \frac{1}{\Delta l_{cc_1}} \Delta \mathbf{u}_{cc_1} \otimes \boldsymbol{\tau}_1, \quad (48a)$$

$$Q_2 = \frac{\mu_2}{\Delta l_{c_2 p}} \Delta \mathbf{u}_{c_2 p} \otimes \mathbf{n}_2 + \frac{1}{\Delta l_{cc_2}} \Delta \mathbf{u}_{cc_2} \otimes \boldsymbol{\tau}_2, \quad (48b)$$

where μ_1 , $\Delta \mathbf{u}_{c_1 p}$, $\Delta \mathbf{u}_{cc_1}$, μ_2 , $\Delta \mathbf{u}_{c_2 p}$, and $\Delta \mathbf{u}_{cc_2}$ are not specified. Therefore, another six equations should be provided for these independent components.

To ensure that the Generalized Riemann Invariant relation holds for this system, the tensor viscosity should degenerate into the scalar viscosity along the edge normal directions. From Cauchy's stress theorem, we obtain

$$|Q_1 \mathbf{n}_1| = |\alpha_1 \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1|, \quad (49a)$$

$$|Q_2 \mathbf{n}_2| = |\alpha_2 \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2|. \quad (49b)$$

It is obvious that Q_1 and Q_2 are zero when $|\Delta \mathbf{u}_{cp}| = 0$. Under the more general condition that $|\Delta \mathbf{u}_{cp}| \neq 0$, recall that Eq. (17) ensures consistency with the GCL. This condition can be obtained with a constant distribution or a linear interpolation of velocity along the edges $[c, c^+]$ and $[c, c^-]$. Note that only $\mathbf{u}_c$ will be determined uniquely through an iterative procedure, whereas $\mathbf{u}_{c^+}$ and $\mathbf{u}_{c^-}$ will not. This makes the constant distribution assumption reasonable. Indeed, if all these velocities were determined uniquely, it might violate the constant distribution assumption.

- Assuming a constant distribution of velocity along the edges $[c, c^+]$ and $[c, c^-]$, which means

$$\Delta \mathbf{u}_{c_1 p} = \Delta \mathbf{u}_{cp}, \quad \Delta \mathbf{u}_{cc_1} = 0, \quad (50a)$$

$$\Delta \mathbf{u}_{c_2 p} = \Delta \mathbf{u}_{cp}, \quad \Delta \mathbf{u}_{cc_2} = 0. \quad (50b)$$

Combining Eqs. (48), (49), and (50), the tensor viscosities are specified as

$$Q_1 = \alpha_1 \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1|}{|\Delta \mathbf{u}_{cp}|} \Delta \mathbf{u}_{cp} \otimes \mathbf{n}_1, \quad (51a)$$

$$Q_2 = \alpha_2 \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2|}{|\Delta \mathbf{u}_{cp}|} \Delta \mathbf{u}_{cp} \otimes \mathbf{n}_2. \quad (51b)$$

The surface integration of these viscosities exactly recovers the general form of M_{cp} in Eq. (36), i.e.,

$$M_{cp} = \left(\alpha_1 l_1 \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1|}{|\Delta \mathbf{u}_{cp}|} + \alpha_2 l_2 \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2|}{|\Delta \mathbf{u}_{cp}|} \right) I, \quad (52)$$

similar to the scalar viscosity, and an iterative procedure is performed to compute $\mathbf{u}_c$. Substituting Eq. (52) into Eq. (29) gives the expression of the subcell force:

$$\mathbf{f}_{cp} = p_c \mathbf{s}_{cp} - \left(\alpha_1 l_1 \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1|}{|\Delta \mathbf{u}_{cp}|} + \alpha_2 l_2 \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2|}{|\Delta \mathbf{u}_{cp}|} \right) \Delta \mathbf{u}_{cp}. \quad (53)$$

- Assuming a linear interpolation of velocity along the edges $[c, c^+]$ and $[c, c^-]$, the following velocity differences are approximated using the vector split method:

$$\Delta \mathbf{u}_{c_1 p} = (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1) \mathbf{n}_1, \quad \Delta \mathbf{u}_{cc_1} = (\Delta \mathbf{u}_{cp} \cdot \boldsymbol{\tau}_1) \boldsymbol{\tau}_1, \quad (54a)$$

$$\Delta \mathbf{u}_{c_2 p} = (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2) \mathbf{n}_2, \quad \Delta \mathbf{u}_{cc_2} = (\Delta \mathbf{u}_{cp} \cdot \boldsymbol{\tau}_2) \boldsymbol{\tau}_2. \quad (54b)$$

Combining Eqs. (48), (49), and (54), the tensor viscosities are specified as

$$Q_1 = \alpha_1 (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1) \mathbf{n}_1 \otimes \mathbf{n}_1 + \frac{\alpha_1 \Delta l_{c_1 p}}{\Delta l_{cc_1}} (\Delta \mathbf{u}_{cp} \cdot \boldsymbol{\tau}_1) \boldsymbol{\tau}_1 \otimes \boldsymbol{\tau}_1, \quad (55a)$$

$$Q_2 = \alpha_2 (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2) \mathbf{n}_2 \otimes \mathbf{n}_2 + \frac{\alpha_2 \Delta l_{c_2 p}}{\Delta l_{cc_2}} (\Delta \mathbf{u}_{cp} \cdot \boldsymbol{\tau}_2) \boldsymbol{\tau}_2 \otimes \boldsymbol{\tau}_2. \quad (55b)$$

The surface integration of these viscosities exactly recovers the general form of M_{cp} in Eq. (36), i.e.,

$$M_{cp} = \alpha_1 l_1 \mathbf{n}_1 \otimes \mathbf{n}_1 + \alpha_2 l_2 \mathbf{n}_2 \otimes \mathbf{n}_2, \quad (56)$$

and the expression of the subcell force is

$$\mathbf{f}_{cp} = p_c \mathbf{s}_{cp} - (\alpha_1 l_1 \mathbf{n}_1 \otimes \mathbf{n}_1 + \alpha_2 l_2 \mathbf{n}_2 \otimes \mathbf{n}_2) \Delta \mathbf{u}_{cp}. \quad (57)$$

It is obvious that the mathematical expressions of $\mathbf{f}_{cp}$ in Eqs. (44) and (57) are exactly the same. Therefore, it can be concluded that the artificial viscosity defined by Eq. (55) is essentially a full tensor form of the scalar artificial viscosity defined by Eq. (40).

Up to now, using different approximations of the velocity distribution, a new tensor artificial viscosity has been developed and the tensor form of the scalar artificial viscosity has been recovered. For simplicity, these are referred to as the tensor viscosity and the scalar viscosity in the remainder of this paper.

The two viscosities can be compared by analyzing the corresponding discrete velocity gradients. Combining Eqs. (47), (50), and (54), the discrete velocity gradients are specified as follows:

$$G_t^+ = \frac{1}{\Delta l_{c_1 p}} \Delta \mathbf{u}_{cp} \otimes \mathbf{n}_1, \quad (58a)$$

$$G_t^- = \frac{1}{\Delta l_{c_1 p}} \Delta \mathbf{u}_{cp} \otimes \mathbf{n}_2, \quad (58b)$$

$$G_s^+ = \frac{1}{\Delta l_{c_1 p}} (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_1) \mathbf{n}_1 \otimes \mathbf{n}_1 + \frac{1}{\Delta l_{cc_1}} (\Delta \mathbf{u}_{cp} \cdot \boldsymbol{\tau}_1) \boldsymbol{\tau}_1 \otimes \boldsymbol{\tau}_1, \quad (58c)$$

$$G_s^- = \frac{1}{\Delta l_{c_2 p}} (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_2) \mathbf{n}_2 \otimes \mathbf{n}_2 + \frac{1}{\Delta l_{cc_2}} (\Delta \mathbf{u}_{cp} \cdot \boldsymbol{\tau}_2) \boldsymbol{\tau}_2 \otimes \boldsymbol{\tau}_2, \quad (58d)$$

where the superscripts “+” and “-” indicate edges $[c, c^+]$ and $[c, c^-]$, respectively, and the subscripts “t” and “s” denote the tensor viscosity and scalar viscosity, respectively. It is obvious that the scalar viscosity requires a symmetric form of the velocity gradient matrix. However, in many cases, the velocity gradient matrix might be asymmetric because of the specific flow structure or the grid strategy. With such a viscosity, Campbell et al. [7] showed that, for a shear flow in which all velocity vectors are parallel, the viscous force would not be parallel to the velocity directions. In contrast, the tensor viscosity is applicable to arbitrary forms of the velocity gradient, both symmetric and asymmetric. Therefore, it is desirable to develop a tensor artificial viscosity that does not require a symmetric velocity gradient matrix.

In the solution algorithm, the only difference between the scalar viscosity and the tensor viscosity is the definition of the subcell matrix M_{cp} . This indicates that the tensor viscosity does not increase the complexity of the algorithm.

3.4. Second-order extension

As the Godunov-type artificial viscosities employ explicit velocity values, the amount of dissipation can be regulated by the reconstruction of the velocity. The second-order extension would involve a linear reconstruction of the velocity.

$$\mathbf{u}_p(\mathbf{x}_c) = \mathbf{u}_p + \nabla \mathbf{u}_p (\mathbf{x}_c - \mathbf{x}_p), \quad (59)$$

where the velocity gradient $\nabla \mathbf{u}_p$ is estimated using Gauss's divergence theorem in the dual cell Ω_p :

$$\nabla \mathbf{u}_p = \frac{1}{V_p} \int_{\partial \Omega_p} \mathbf{u} \cdot \mathbf{n} dS. \quad (60)$$

Here, V_p denotes the volume of Ω_p . The velocity in each integration path is calculated as the average of the endpoint velocities. To preserve monotonicity, a vector limiter is required [32], we follow the approach of Maire et al. to develop the procedure. With this limiter, the reconstructed velocity will not exceed the maximum and minimum velocity values located at the vertices of Ω_p . Consider a local coordinate system whose basis vectors are $\mathbf{e}_1$ and $\mathbf{e}_2$. We set $\mathbf{e}_p^\parallel$ to be parallel to $\mathbf{u}_p$, and $\mathbf{e}_p^\perp$ to be perpendicular to $\mathbf{e}_p^\parallel$. Therefore, a transformation matrix $A_p = (\mathbf{e}_p^\parallel, \mathbf{e}_p^\perp)$ from the original coordinate system to the local coordinate system is defined. The vector limiter is specified as follows:

$$\boldsymbol{\phi}_p = A_p^{-1} \begin{pmatrix} \phi_p^\parallel & 0 \\ 0 & \phi_p^\perp \end{pmatrix} A_p, \quad (61)$$

where $\phi_p^\parallel$ and $\phi_p^\perp$ are calculated using the Barth-Jespersen method [33] when the velocity field is projected into the directions of $\mathbf{e}_p^\parallel$ and $\mathbf{e}_p^\perp$, respectively. The idea of the Barth-Jespersen method is to find the largest admissible limiter to preserve monotonicity. In practice, we make the limiter more flexible by multiplying a coefficient $0 \leq \beta \leq 1$; β is set to 1 if not specified. For the second-order extension, the following limited reconstruction of the velocity is performed:

$$\mathbf{u}_p(\mathbf{x}_c) = \mathbf{u}_p + \beta \boldsymbol{\phi}_p \nabla \mathbf{u}_p (\mathbf{x}_c - \mathbf{x}_p). \quad (62)$$

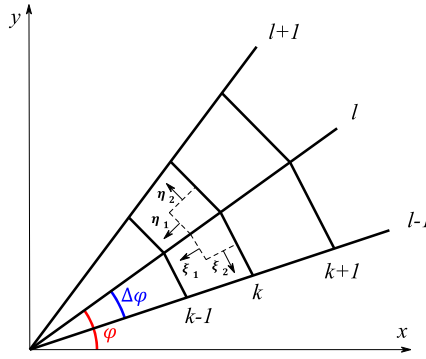


Fig. 3. Schematic of the equi-angular polar grid.

4. Symmetry property of the tensor artificial viscosity

In this section, the developed tensor artificial viscosity is shown to maintain 1D cylindrical symmetry on an equi-angular initial grid.

Cylindrical symmetry of the tensor artificial viscosity should consider variables including the momentum, internal energy, and density. For an equi-angular grid, the variable values should be identical along the angular directions, and the momentum should be equal along the radial direction.

Theorem. For an equi-angular initial grid, the tensor artificial viscosity preserves the 1D cylindrical symmetry property of the solution to Eq. (24) as the discrete time steps progress.

Proof. Without loss of generality, it is only necessary to prove that, when the momentum, internal energy, and density are known to have cylindrical symmetry at the n -th step, these variables maintain cylindrical symmetry at the $(n + \frac{1}{2})$ -th step.

As depicted in Fig. 3, the 2D domain Ω is discretized into $K \times L$ equi-angular cells. The indices along the radial and angular directions are k and l , respectively. The angle between the l -th line and the x axis is φ , and the angle between any two neighboring angular lines is $\Delta\varphi$. For the cell $I_{k-\frac{1}{2},l-\frac{1}{2}}$, $I_{k-\frac{1}{2},l+\frac{1}{2}}$, $I_{k+\frac{1}{2},l+\frac{1}{2}}$, and $I_{k+\frac{1}{2},l-\frac{1}{2}}$, let us connect the cell centroids with the midpoints of edges who share the same vertex in sequence. This operation creates the dual grid $J_{k,l}$, in which the momentum is defined as

$$\mathbf{M}_{k,l}^{n+\frac{1}{2}} = \Delta t (\mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}} + \mathbf{f}_{k-\frac{1}{2},l+\frac{1}{2}} + \mathbf{f}_{k+\frac{1}{2},l+\frac{1}{2}} + \mathbf{f}_{k+\frac{1}{2},l-\frac{1}{2}}), \quad (63)$$

where Δt is the time step from the n -th step to the $(n+1)$ -th step, $\mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}}$ denotes the subcell force acting on cell $J_{k,l}$ from cell $I_{k-\frac{1}{2},l-\frac{1}{2}}$, and $\mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}}$ only considers the viscous term.

Firstly, we prove the cylindrical symmetry of $(\mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}} + \mathbf{f}_{k-\frac{1}{2},l+\frac{1}{2}})$. The unit normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ defined in Eq. (38) are specified as ξ_1 and ξ_2 for $I_{k-\frac{1}{2},l-\frac{1}{2}}$. By analogy, η_1 and η_2 are defined for $I_{k-\frac{1}{2},l+\frac{1}{2}}$.

$$\xi_1 = \begin{pmatrix} -\cos(\varphi - \frac{1}{2}\Delta\varphi) \\ -\sin(\varphi - \frac{1}{2}\Delta\varphi) \end{pmatrix}, \quad \xi_2 = \begin{pmatrix} \sin(\varphi - \frac{1}{2}\Delta\varphi) \\ -\cos(\varphi - \frac{1}{2}\Delta\varphi) \end{pmatrix}, \quad (64a)$$

$$\eta_1 = \begin{pmatrix} -\cos(\varphi + \frac{1}{2}\Delta\varphi) \\ -\sin(\varphi + \frac{1}{2}\Delta\varphi) \end{pmatrix}, \quad \eta_2 = \begin{pmatrix} -\sin(\varphi + \frac{1}{2}\Delta\varphi) \\ \cos(\varphi + \frac{1}{2}\Delta\varphi) \end{pmatrix}. \quad (64b)$$

In Fig. 3, the lengths of the dashed edges with the unit normal vectors ξ_1 and ξ_2 are $\Delta l_{k-\frac{1}{2},l-\frac{1}{2}}$ and $\Delta d_{k-\frac{1}{2},l-\frac{1}{2}}$, respectively.

As the conserved variables at the n -th step have cylindrical symmetry, their values are identical when they have the same index k or $k - 1/2$. The following simplified symbols are used for convenience:

$$\begin{aligned} m_{k,l} &= m_k, & |\mathbf{u}_{k,l}| &= u_k, & |\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}}| &= u_{k-\frac{1}{2}}, \\ \rho_{k-\frac{1}{2},l-\frac{1}{2}} &= \rho, & b_{k-\frac{1}{2},l-\frac{1}{2}} &= b, & a_{k-\frac{1}{2},l-\frac{1}{2}} &= a, \\ \Delta l_{k-\frac{1}{2},l-\frac{1}{2}} &= \Delta l, & \Delta d_{k-\frac{1}{2},l-\frac{1}{2}} &= \Delta d, \\ m_{k-\frac{1}{2},l-\frac{1}{2}} &= m_{k-\frac{1}{2}}, & \varepsilon_{k-\frac{1}{2},l-\frac{1}{2}} &= \varepsilon_{k-\frac{1}{2}}, \end{aligned} \quad (65)$$

where $a_{k-\frac{1}{2},l-\frac{1}{2}}$ is the speed of sound in cell $I_{k-\frac{1}{2},l-\frac{1}{2}}$, and $b_{k-\frac{1}{2},l-\frac{1}{2}}$ represents the parameter in cell $I_{k-\frac{1}{2},l-\frac{1}{2}}$ in Eq. (39).

The velocities at vertex (k, l) and cell center $(k - 1/2, l - 1/2)$ are

$$\mathbf{u}_{k,l} = \begin{pmatrix} u_k \cos \varphi \\ u_k \sin \varphi \end{pmatrix}, \quad \mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}} = \begin{pmatrix} u_{k-\frac{1}{2}} \cos(\varphi - \frac{1}{2}\Delta\varphi) \\ u_{k-\frac{1}{2}} \sin(\varphi - \frac{1}{2}\Delta\varphi) \end{pmatrix}. \quad (66)$$

For convenience, the symbol $\Delta\mathbf{u}$ represents the velocity jump in the Riemann problem, i.e.,

$$\Delta\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}} = \mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}} - \mathbf{u}_{k,l}. \quad (67)$$

According to Eq. (53), the subcell force from $I_{k-\frac{1}{2},l-\frac{1}{2}}$ acting on $J_{k,l}$ is

$$\begin{aligned} \mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}} &= \left[\Delta d \rho \left(a + b |\Delta\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}} \cdot \boldsymbol{\xi}_2| \right) \frac{|\Delta\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}} \cdot \boldsymbol{\xi}_2|}{|\Delta\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}}|} \right. \\ &\quad \left. + \Delta l \rho \left(a + b |\Delta\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}} \cdot \boldsymbol{\xi}_1| \right) \frac{|\Delta\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}} \cdot \boldsymbol{\xi}_1|}{|\Delta\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}}|} \right] \cdot \Delta\mathbf{u}_{k-\frac{1}{2},l-\frac{1}{2}} \\ &= -\frac{\rho}{|\Delta u_{k-\frac{1}{2}}|} \left[\Delta d \left(a + b \left| u_k \sin \left(\frac{1}{2} \Delta\varphi \right) \right| \right) + \right. \\ &\quad \left. \Delta l \left(a + b \left| -u_{k-\frac{1}{2}} + u_k \cos \left(\frac{1}{2} \Delta\varphi \right) \right| \right) \right] \begin{pmatrix} u_{k-\frac{1}{2}} \cos(\varphi - \frac{1}{2}\Delta\varphi) - u_k \cos \varphi \\ u_{k-\frac{1}{2}} \sin(\varphi - \frac{1}{2}\Delta\varphi) - u_k \sin \varphi \end{pmatrix}. \end{aligned} \quad (68)$$

Similarly, the subcell force from $I_{k-\frac{1}{2},l+\frac{1}{2}}$ acting on $J_{k,l}$ is

$$\begin{aligned} \mathbf{f}_{k-\frac{1}{2},l+\frac{1}{2}} &= -\frac{\rho}{|\Delta u_{k-\frac{1}{2}}|} \left[\Delta d \left(a + b \left| u_k \sin \left(\frac{1}{2} \Delta\varphi \right) \right| \right) + \right. \\ &\quad \left. \Delta l \left(a + b \left| -u_{k-\frac{1}{2}} + u_k \cos \left(\frac{1}{2} \Delta\varphi \right) \right| \right) \right] \begin{pmatrix} u_{k-\frac{1}{2}} \cos(\varphi + \frac{1}{2}\Delta\varphi) - u_k \cos \varphi \\ u_{k-\frac{1}{2}} \sin(\varphi + \frac{1}{2}\Delta\varphi) - u_k \sin \varphi \end{pmatrix}. \end{aligned} \quad (69)$$

The summation of $\mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}}$ and $\mathbf{f}_{k-\frac{1}{2},l+\frac{1}{2}}$ is

$$\begin{aligned} \mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}} + \mathbf{f}_{k-\frac{1}{2},l+\frac{1}{2}} &= - \left\{ \frac{2\rho(u_{k-\frac{1}{2}} \cos(\frac{1}{2}\Delta\varphi) - u_k)}{|\Delta u_{k-\frac{1}{2}}|} \right. \\ &\quad \left. \left[\Delta d \left(a + b \left| -u_{k-\frac{1}{2}} + u_k \cos \left(\frac{1}{2} \Delta\varphi \right) \right| \right) + \Delta l \left(a + b \left| u_k \sin \left(\frac{1}{2} \Delta\varphi \right) \right| \right) \right] \right\} \begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix}. \end{aligned} \quad (70)$$

It is obvious that the vector component of $(\mathbf{f}_{k+\frac{1}{2},l+\frac{1}{2}} + \mathbf{f}_{k+\frac{1}{2},l-\frac{1}{2}})$ is $(\cos \varphi, \sin \varphi)^T$, which represents the direction of the l -th line. In addition, Δd , Δl , and $\Delta\varphi$ are independent of the index l . Thus, the absolute values of $(\mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}} + \mathbf{f}_{k-\frac{1}{2},l+\frac{1}{2}})$ are identical for the same index value k . A similar conclusion can be drawn for $(\mathbf{f}_{k+\frac{1}{2},l+\frac{1}{2}} + \mathbf{f}_{k+\frac{1}{2},l-\frac{1}{2}})$. Thus, the resulting $\mathbf{M}_{k,l}^{n+\frac{1}{2}}$ from Eq. (63) preserves the property of cylindrical symmetry.

Analysis of the internal energy symmetry starts with Eq. (30b). The internal energy of cell $I_{k-\frac{1}{2},l-\frac{1}{2}}$ at time step $n + \frac{1}{2}$ can be expressed as

$$\varepsilon_{k-\frac{1}{2},l-\frac{1}{2}}^{n+\frac{1}{2}} = \varepsilon_{k-\frac{1}{2}} + \frac{\Delta t}{m_{k-\frac{1}{2}}} (w_{k-1,l-1} + w_{k,l-1} + w_{k,l} + w_{k-1,l}). \quad (71)$$

For subcell $I_{k-\frac{1}{2},l-\frac{1}{2}} \cap J_{k,l}$ at t_n , the symbol $w_{k,l}$ is computed as the product of $\mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}}$ and $\mathbf{u}_{k,l}$:

$$\begin{aligned} w_{k,l} &= \mathbf{f}_{k-\frac{1}{2},l-\frac{1}{2}} \cdot \mathbf{u}_{k,l} \\ &= -\frac{\rho}{|\Delta u_{k-\frac{1}{2}}|} \left[\Delta d \left(a + b \left| u_k \sin \left(\frac{1}{2} \Delta\varphi \right) \right| \right) + \right. \\ &\quad \left. \Delta l \left(a + b \left| -u_{k-\frac{1}{2}} + u_k \cos \left(\frac{1}{2} \Delta\varphi \right) \right| \right) \right] \left[u_k u_{k-\frac{1}{2}} \cos \left(\frac{1}{2} \Delta\varphi \right) - u_k^2 \right]. \end{aligned} \quad (72)$$

It is obvious the absolute value of $w_{k,l}$ is identical for the same index k . Similar conclusions can be drawn for $w_{k-1,l}$, $w_{k,l-1}$, and $w_{k-1,l-1}$. Therefore, the internal energy preserves the property of cylindrical symmetry.

Based on the symmetry of momentum, it is easy to prove that the velocity $\mathbf{u}_{k,l}^{n+\frac{1}{2}}$ and the vertex position $\mathbf{x}_{k,l}^{n+\frac{1}{2}}$ also preserve cylindrical symmetry. Their expressions can be defined as follows:

$$\mathbf{u}_{k,l}^{n+\frac{1}{2}} = \begin{pmatrix} u_k^{n+\frac{1}{2}} \cos \varphi \\ u_k^{n+\frac{1}{2}} \sin \varphi \end{pmatrix}, \quad \mathbf{x}_{k,l}^{n+\frac{1}{2}} = \begin{pmatrix} x_k^{n+\frac{1}{2}} \cos \varphi \\ x_k^{n+\frac{1}{2}} \sin \varphi \end{pmatrix}. \quad (73)$$

Define two triangles whose vertex positions are $\{0, \mathbf{x}_{k,l-1}^{n+\frac{1}{2}}, \mathbf{x}_{k,l}^{n+\frac{1}{2}}\}$ and $\{0, \mathbf{x}_{k-1,l-1}^{n+\frac{1}{2}}, \mathbf{x}_{k-1,l}^{n+\frac{1}{2}}\}$, respectively. The volume of cell $I_{k-\frac{1}{2},l-\frac{1}{2}}$ at time step $n + \frac{1}{2}$ can be calculated as the difference in the triangle volumes, i.e.,

$$V_{k-\frac{1}{2},l-\frac{1}{2}}^{n+\frac{1}{2}} = \left[\left(x_k^{n+\frac{1}{2}} \right)^2 - \left(x_{k-1}^{n+\frac{1}{2}} \right)^2 \right] \sin \left(\frac{1}{2} \Delta \varphi \right) \cos \left(\frac{1}{2} \Delta \varphi \right). \quad (74)$$

The density of cell $I_{k-\frac{1}{2},l-\frac{1}{2}}$ at time step $n + \frac{1}{2}$ is

$$\rho_{k-\frac{1}{2},l-\frac{1}{2}}^{n+\frac{1}{2}} = \frac{m_{k-\frac{1}{2}}}{\left[\left(x_k^{n+\frac{1}{2}} \right)^2 - \left(x_{k-1}^{n+\frac{1}{2}} \right)^2 \right] \sin \left(\frac{1}{2} \Delta \varphi \right) \cos \left(\frac{1}{2} \Delta \varphi \right)}. \quad (75)$$

Thus, the cell density is independent of index l , and preserves the property of cylindrical symmetry. The above analysis completes the full proof of cylindrical symmetry. $\square$

5. Numerical examples

A series of numerical simulations were conducted with the ideal gas EOS. The tensor artificial viscosity was applied to Sod's problem, the Taylor–Green vortex, the Rayleigh–Taylor instability, and the Sedov and Saltzman problems to test its numerical accuracy. The Noh and triple-point problems were also examined to demonstrate the anti-hourglass feature of the tensor artificial viscosity.

In the numerical tests, the scalar artificial viscosity and the tensor artificial viscosity are both implemented for comparison. The temporary quadrilateral sub-zoning technique [34] is available for the anti-hourglass mode. However, it is not adopted in the current simulations to prevent the resulting uncertainties.

5.1. Sod's problem

Sod's problem [35] consists of a two-material discontinuity in a shock tube of unit length. Initially, the discontinuity is located at $x = 0.5$. The left and right side states are characterized by $\{\rho_L, P_L, u_L, \gamma_L\} = \{1, 1, 0, 1.4\}$ and $\{\rho_R, P_R, u_R, \gamma_R\} = \{0.125, 0.1, 0, 1.4\}$, respectively. The computational domain is defined by $(x, y) \in [0, 1] \times [0, 0.01]$. A 2D Cartesian grid that is equally divided in both the x and y directions is used. The mesh resolution is 100×4 , and the boundary is a reflective wall, where the normal velocity is 0 at the boundary.

The simulated results of density, pressure, and internal energy at $t = 0.2$ with first- and second-order spatial accuracy are presented in Fig. 4. The main conclusions are as follows. The results are convergent, and the second-order scheme gives more accurate results than the first-order scheme. Moreover, the numerical results for the tensor artificial viscosity are identical to those for the scalar artificial viscosity, which conforms to Eq. (45) in one dimension.

5.2. Taylor–Green vortex problem

This case [21] describes a smooth steady flow problem with vorticity. Usually, the simulation only runs for a short time because large deformations occur with time. Consider the compressible Euler equations characterized by the following initial conditions:

$$\mathbf{u}_0 = \{\sin(\pi x) \cos(\pi y), -\cos(\pi x) \sin(\pi y)\}, \quad (76a)$$

$$\rho_0 = 1, \quad (76b)$$

$$p_0 = \frac{1}{4} \rho_0 [\cos(2\pi x) + \cos(2\pi y)] + 1, \quad (76c)$$

$$\varepsilon_0 = \frac{p_0}{\rho_0(\gamma - 1)}. \quad (76d)$$

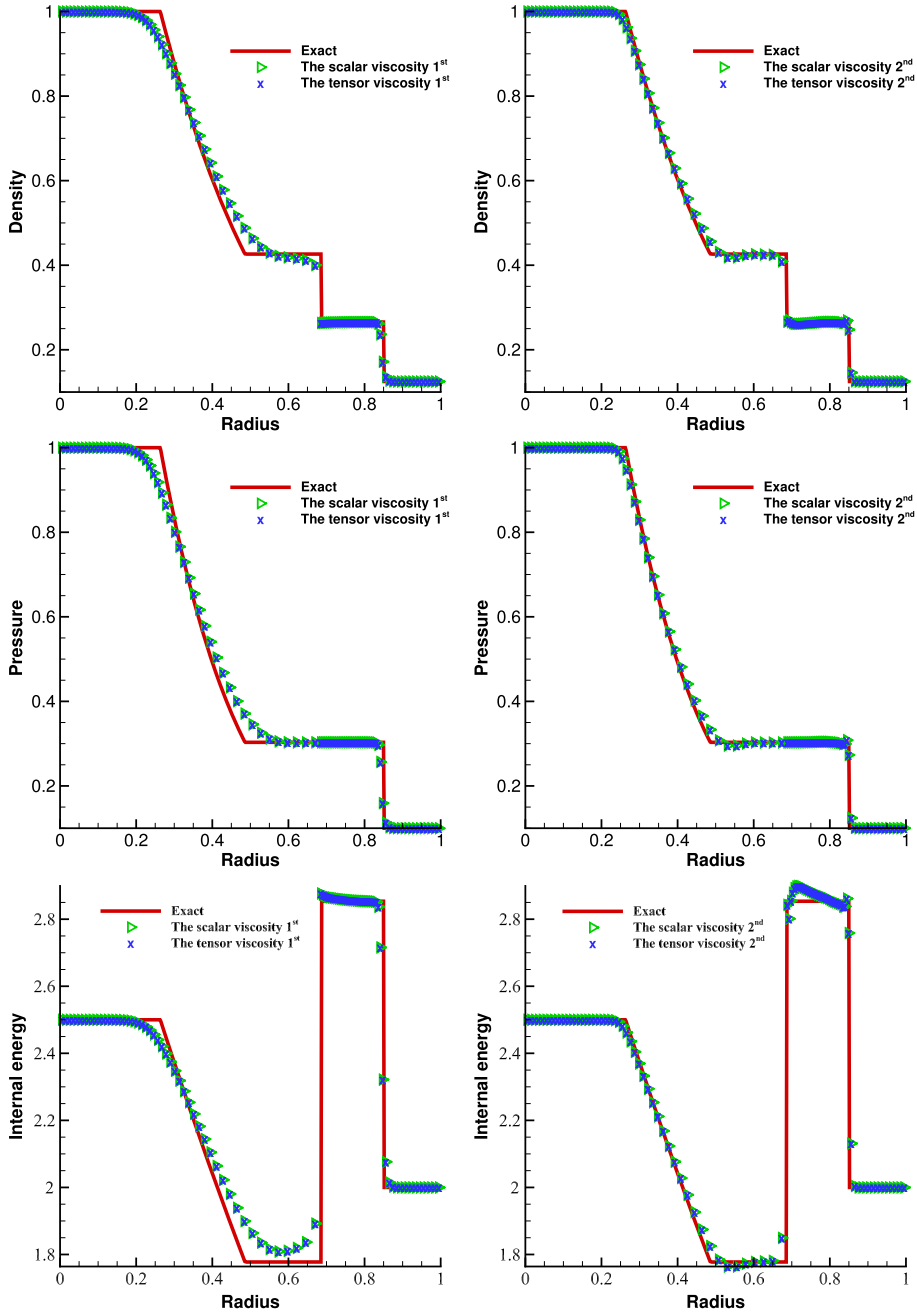


Fig. 4. Simulation results for Sod's problem at $t = 0.2$ with 100×4 Cartesian grid cells. From top to bottom: density, pressure, and internal energy. Left: first-order scheme; right: second-order scheme.

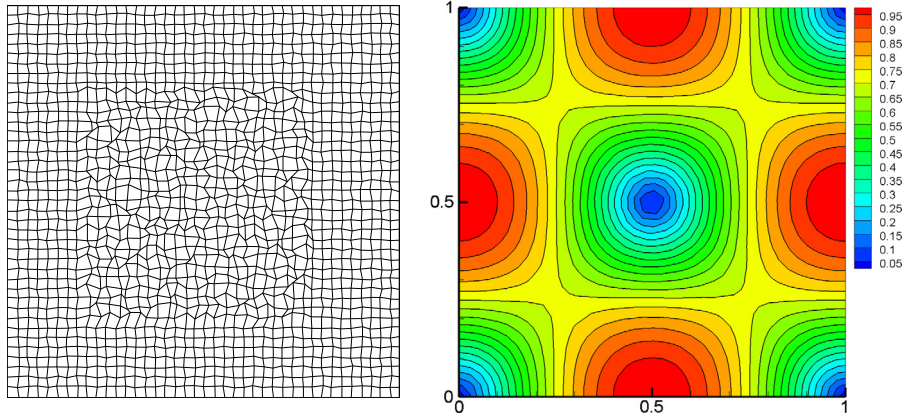
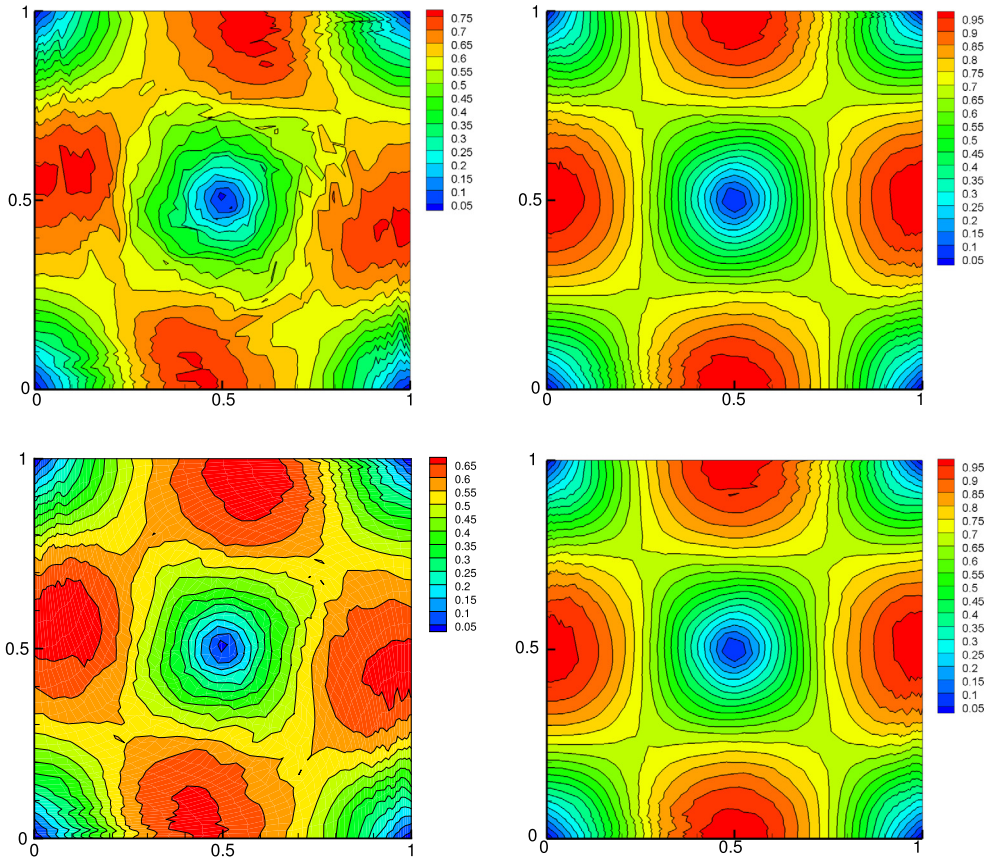
The computational domain is $[0, 1] \times [0, 1]$ with wall boundary conditions. The ideal gas EOS is applied with $\gamma = 1.4$. An extra source term is added on the right-hand side of the internal equation (23) to ensure the flow remains steady:

$$S_e = \frac{5\pi}{8} [\cos(\pi x) \cos(3\pi y) - \cos(3\pi x) \cos(\pi y)]. \quad (77)$$

Firstly, simulations were performed on uniform Cartesian grids to analyze the convergence rate of the numerical scheme. The results at resolutions of 20×20 , 40×40 , 80×80 , and 160×160 cells are presented in Table 2. Note that for both artificial viscosities, the expected convergence rates are observed for the first- and second-order schemes. Secondly, the simulation of a randomly distorted 40×40 grid was considered. The initial grid and the exact velocity field are shown in Fig. 5. The simulated velocity fields are shown in Fig. 6. For the first-order scheme, the tensor viscosity generates a smoother

Table 2 L_2 error and convergence rate at $t = 0.6$ with uniform Cartesian grids.

Cells	First-order				Second-order			
	scalar		tensor		scalar		tensor	
	L_2 error	rate	L_2 error	rate	L_2 error	rate	L_2 error	rate
20×20	2.460E-1	–	3.211E-1	–	4.248E-2	–	3.288E-2	–
40×40	1.355E-1	0.86	1.860E-1	0.79	9.606E-3	2.14	8.722E-3	1.89
80×80	7.120E-2	0.93	1.009E-1	0.88	2.886E-3	1.74	2.488E-3	1.81
160×160	3.658E-2	0.96	5.275E-2	0.94	8.719E-4	1.73	7.938E-4	1.65

**Fig. 5.** Left: a randomly distorted 40×40 grid; right: the exact velocity field.**Fig. 6.** Simulated velocity fields. Top: scalar viscosity; bottom: tensor viscosity; left: first-order scheme; right: second-order scheme.

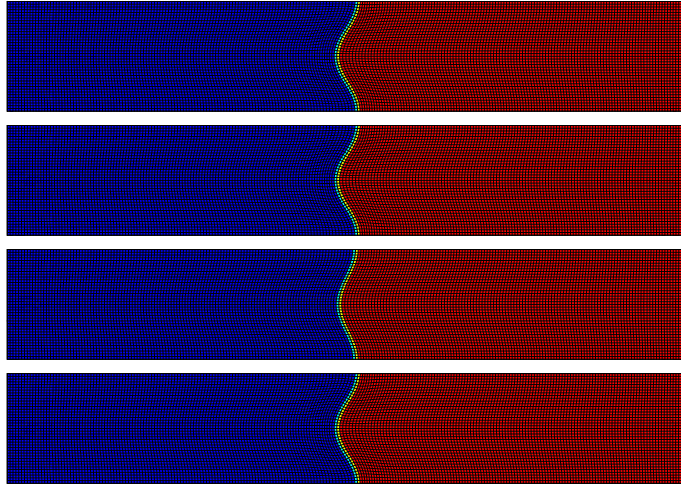


Fig. 7. Density field of the Rayleigh–Taylor instability at time $t = 2.5$. From top to bottom: first-order with scalar viscosity; second-order with scalar viscosity; first-order with tensor viscosity; second-order with tensor viscosity.

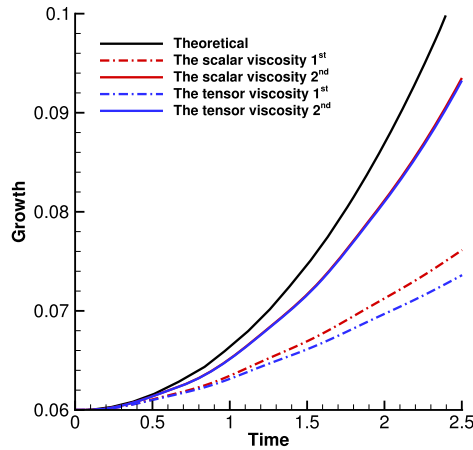


Fig. 8. Growth (computed as the difference in the y coordinate of the lowest fragment of the heavy gas and the highest fragment of the light gas) in the early stages of the Rayleigh–Taylor instability problem.

velocity field than the scalar viscosity, at the cost of increased dissipation. For the second-order scheme, there is a small difference between the scalar and tensor viscosity results. The velocity fields of the second-order scheme exhibit better consistency with the exact solution when compared to the first-order scheme.

5.3. Rayleigh–Taylor instability

The purpose of this example is to assess the growth of the instability in the early stages. Initially, two ideal gases with densities $\rho_h = 2$ and $\rho_l = 1$ are separated by a perturbed interface. The heavy gas is above the light gas. For both gases the adiabatic constant $\gamma = 1.4$, and the interface is described as $y_{dis} = 3 + 0.06 \cos(2\pi x)$. A gravity field acting vertically downward with magnitude $g = 0.1$ is applied. The computational domain is $[0, 1] \times [0, 6]$, and is discretized by 40×240 cells. The initial gas velocities are zero, and the initial gas pressure distribution is

$$p = \begin{cases} 1 + \rho_h g(y - 6), & y > y_{dis}, \\ 1 + \rho_h g(y_{dis} - 6) + \rho_l g(y - y_{dis}), & y < y_{dis}. \end{cases} \quad (78)$$

Fig. 7 shows the density field at time $t = 2.5$. The instability growth before this time can be estimated using the linear theory. Fig. 8 compares the theoretical growth and the numerical growth. It can be observed that the first-order scheme kills the development of this growth for both viscosities, with the result using the tensor viscosity being worse than that using the scalar viscosity. For the second-order scheme, the scalar viscosity and the tensor viscosity produce similar instability growth that is slightly below the theoretical value. These results indicate the ability of the tensor viscosity to simulate problems that involve high vorticity.

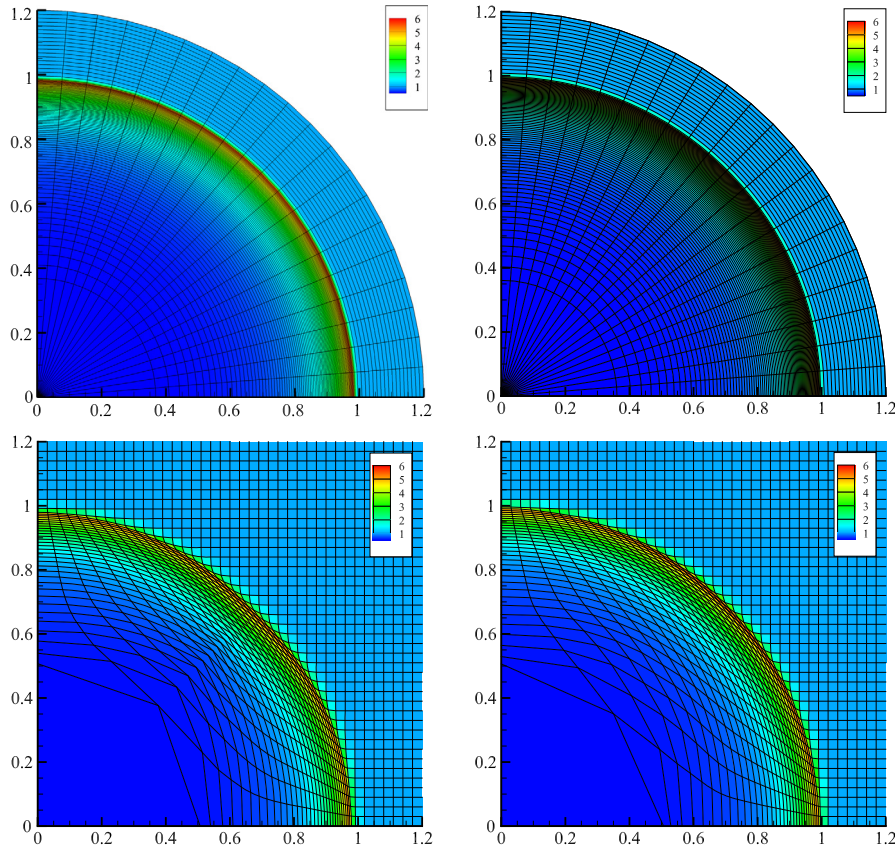


Fig. 9. Sedov's problem using tensor viscosity at $t = 1$. Top: density fields for the 100×20 polar grid; bottom: density fields for the 40×40 Cartesian grid. Left: first-order scheme; right: second-order scheme.

5.4. Sedov's problem

Sedov's problem considers a flow field in which a blast wave evolves from a point explosion. The initial condition of a gas with $\gamma = 1.4$ is $\{\rho_0, P_0, \mathbf{u}_0\} = \{1, 10^{-6}, 0\}$. The initial computation domain is a quarter-circle defined in the region $[0, 1.2] \times [0, 1.2]$. The cells connected with the origin share a total internal energy of 0.244816, and the pressures of these cells are prescribed with the EOS. At $t = 1$, the explosion produces a shock located at a radius of 1 with a density peak of $\rho = 6$. The exact solution to this problem is available in [36]. Two grids are used in this problem: the 100×20 equi-angular polar grid and the 40×40 uniform Cartesian grid.

The results with the tensor viscosity are shown in Figs. 9 and 10. It is clear from Fig. 9 that the grid distribution preserves good 1D symmetry. The density profiles in Fig. 10 indicate good convergence to the exact solution for both the first- and second-order schemes. There are only small differences between the results using scalar viscosity and those using tensor viscosity. Thus, the results produced using scalar viscosity are not presented here. Readers are referred to [26] for a description of the performance of scalar viscosity applied to Sedov's problem.

5.5. Saltzman's problem

Saltzman's problem [37] is known to be a difficult numerical test. It consists of a piston-driven shock that is not aligned with the mesh. The computational domain is $[1, 0.1]$, and this is divided into a 100×10 grid with the following distribution:

$$x = (i - 1)\Delta x + (11 - j)\Delta y \sin(\Delta x(i - 1)\pi), \quad y = (j - 1)\Delta y, \quad (79)$$

where $\Delta x = \Delta y = 0.01$. In the flow field, a piston on the left boundary moves to the right with a constant velocity of 1, generating a shock with a velocity of $4/3$ moving from left to right. The initial condition is $\{\rho_0, P_0, \mathbf{u}_0\} = \{1, 10^{-6}, 0\}$. The right, upper, and lower boundaries are all reflective walls.

Fig. 11 shows the results produced using tensor viscosity before the first shock reflection at $t = 0.6$. The exact shock position is $x = 0.8$, and the simulated results show reasonable consistency with the exact solution for both the first- and second-order schemes. A comparison between the two viscosities are made for the first-order scheme. It is shown in Fig. 12 that, using the scalar artificial viscosity, the calculation cannot continue past $t = 0.9$. However, using the tensor artificial

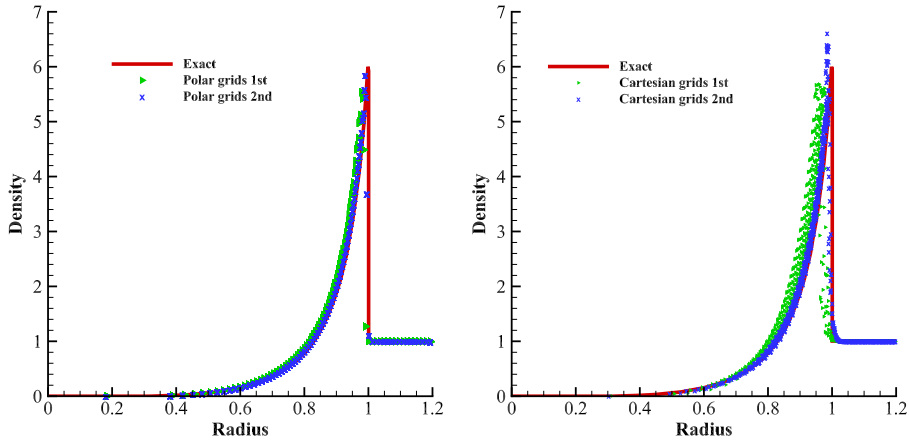


Fig. 10. Density-radius profile of Sedov's problem using tensor viscosity at $t = 1$. Left: first-order scheme; right: second-order scheme.

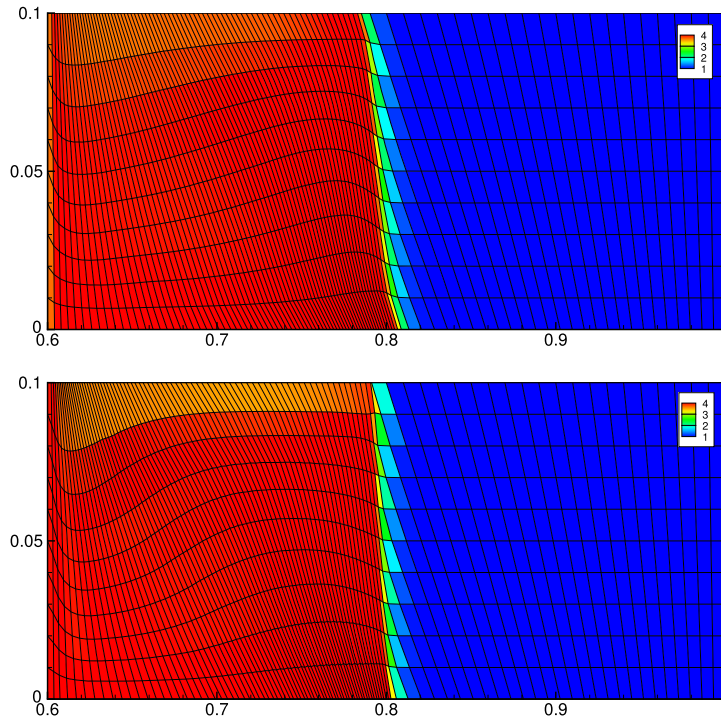


Fig. 11. Density field of Saltzman's problem at $t = 0.6$ using the tensor viscosity. Top: first-order scheme; bottom: second-order scheme.

viscosity, the calculation can run until $t = 0.94$. For the second-order scheme, the simulation using scalar viscosity cannot continue past $t = 0.25$. Thus, the comparisons between the two types of viscosity consider the first-order scheme. Obviously, the numerical scheme with the tensor artificial viscosity gives better results in this case.

5.6. Noh's problem

Noh's problem [38] is a well-known test for validating the Lagrangian scheme in simulations with strong shock waves. The initial conditions are $\{\rho_0, P_0\} = \{1, 10^{-6}\}$, and the unit inward velocity is $\mathbf{u} = (-x/r, -y/r)$. In Noh's problem, a shock wave generated at $(0, 0)$ at the initial time begins to propagate outward. The density behind the shock has a value of 16. At $t = 0.6$, the shock wave reaches a radial coordinate of 0.2. The computation domain is $[0, 1] \times [0, 1]$, and a uniform Cartesian grid is used.

Fig. 13 presents the density maps given by the first-order scheme at $t = 0.6$. In the initial 40×40 Cartesian grid, it is obvious that the mesh preserves good quality solutions with both artificial viscosities. In the initial 30×60 Cartesian grid, some distortions appear in the mesh when using the scalar viscosity, whereas good solutions are given by the tensor

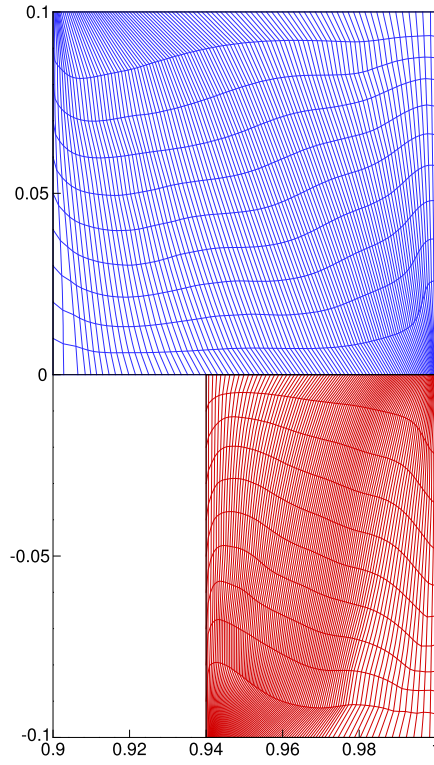


Fig. 12. Mesh of Saltzman's problem in the limiting configuration for the first-order scheme. Top: scalar viscosity; bottom: tensor viscosity.

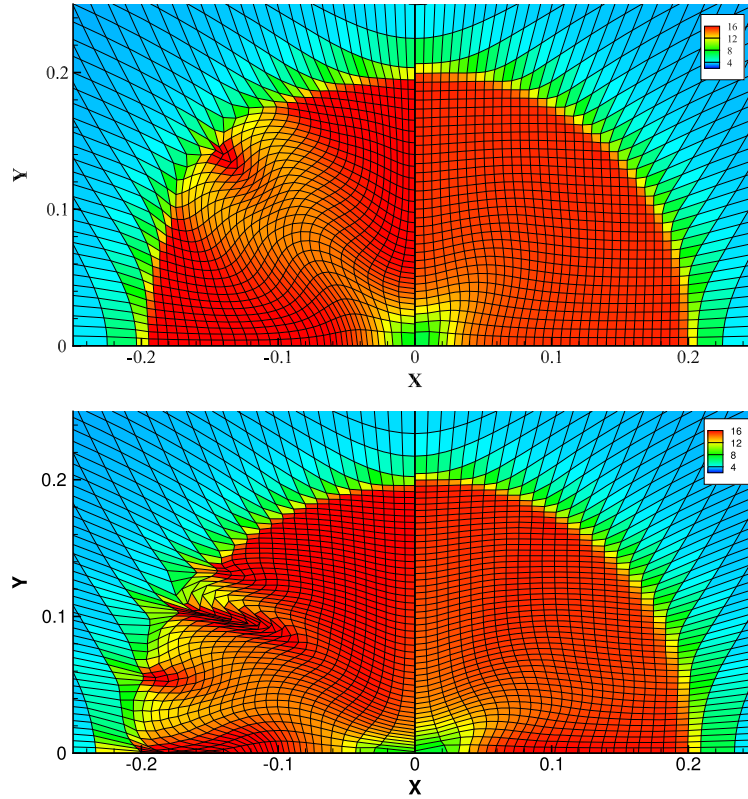


Fig. 13. Density maps of Noh's problem at $t = 0.6$ for the first-order scheme. $X < 0$: scalar viscosity; $X > 0$: tensor viscosity. Top: initial 40×40 Cartesian grid; bottom: initial 30×60 Cartesian grid.

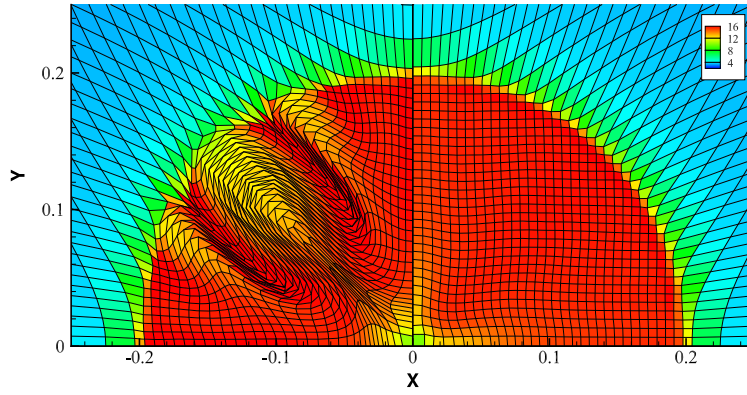


Fig. 14. Density maps of Noh's problem at $t = 0.6$ for the second-order scheme with $\beta = 0.8$ using an initial 40×40 Cartesian grid. $X < 0$: scalar viscosity; $X > 0$: tensor viscosity.

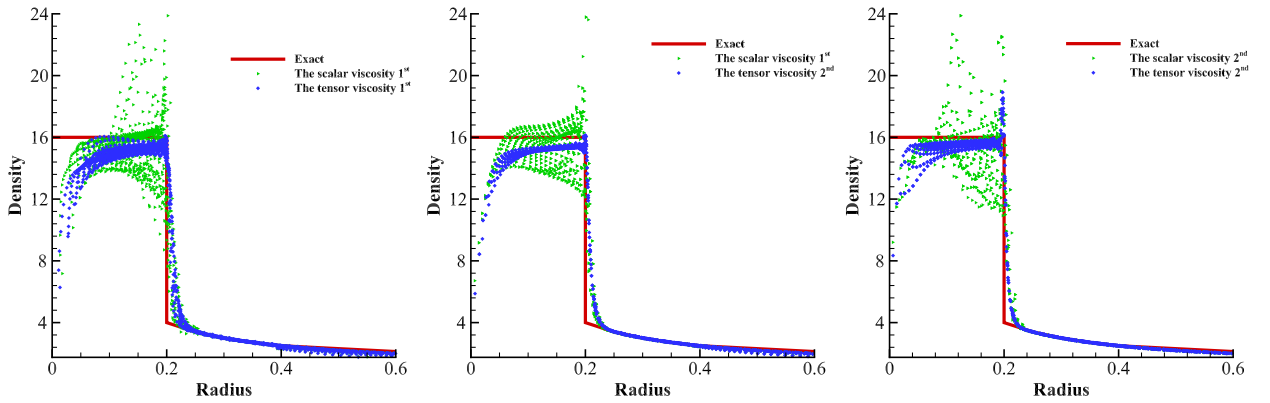


Fig. 15. Density plot of Noh's problem at $t = 0.6$. Left: first-order scheme, initial 30×60 Cartesian grid; middle: first-order scheme, initial 40×40 Cartesian grid; right: second-order scheme with $\beta = 0.8$, initial 40×40 Cartesian grid.

$\rho_1 = 1$ $p_1 = 1$ $\gamma_1 = 1.5$	$\rho_3 = 0.125$ $p_3 = 0.1$ $\gamma_3 = 1.5$
	$\rho_2 = 1$ $p_2 = 0.1$ $\gamma_2 = 1.4$

Fig. 16. Initial conditions of the triple-point problem.

viscosity. For the second-order scheme, the coefficient of the limiter is set to be less than 1 so that the simulations could run until $t = 0.6$. Fig. 14 presents the density maps with $\beta = 0.8$ when $t = 0.6$. In the initial 40×40 Cartesian grid, the mesh becomes distorted in the vicinity of the line $X = -Y$ when using the scalar viscosity. Almost all of the problems disappear when using the tensor viscosity. The results on the initial 30×60 Cartesian grid are not presented, because the simulations of the second-order scheme were unable to reach $t = 0.6$ with either viscosity.

A more detailed analysis using Fig. 15 indicates that the density curves are not convergent for the scalar viscosity, but do converge with the tensor viscosity. Furthermore, in the initial 40×40 Cartesian grid, the density curve of the second-order scheme not only converges, but also shows better consistency with the analytical solution than the first-order scheme.

5.7. Triple-point problem

The triple-point problem with significant physical vortices can be used to assess the robustness of Lagrangian methods [39]. Fig. 16 shows the initial conditions of this problem, and the boundaries are all set as wall conditions. A 70×30 uniform Cartesian grid is used. The resulting meshes and specific internal energy maps at $t = 4.5$ are shown in Fig. 17.

For the first-order scheme, the mesh exhibits apparent distortion under the scalar viscosity, but remains stable with the tensor viscosity. As the dissipation of the scalar viscosity decreases, the vortex rolls up, at the cost of more severe mesh

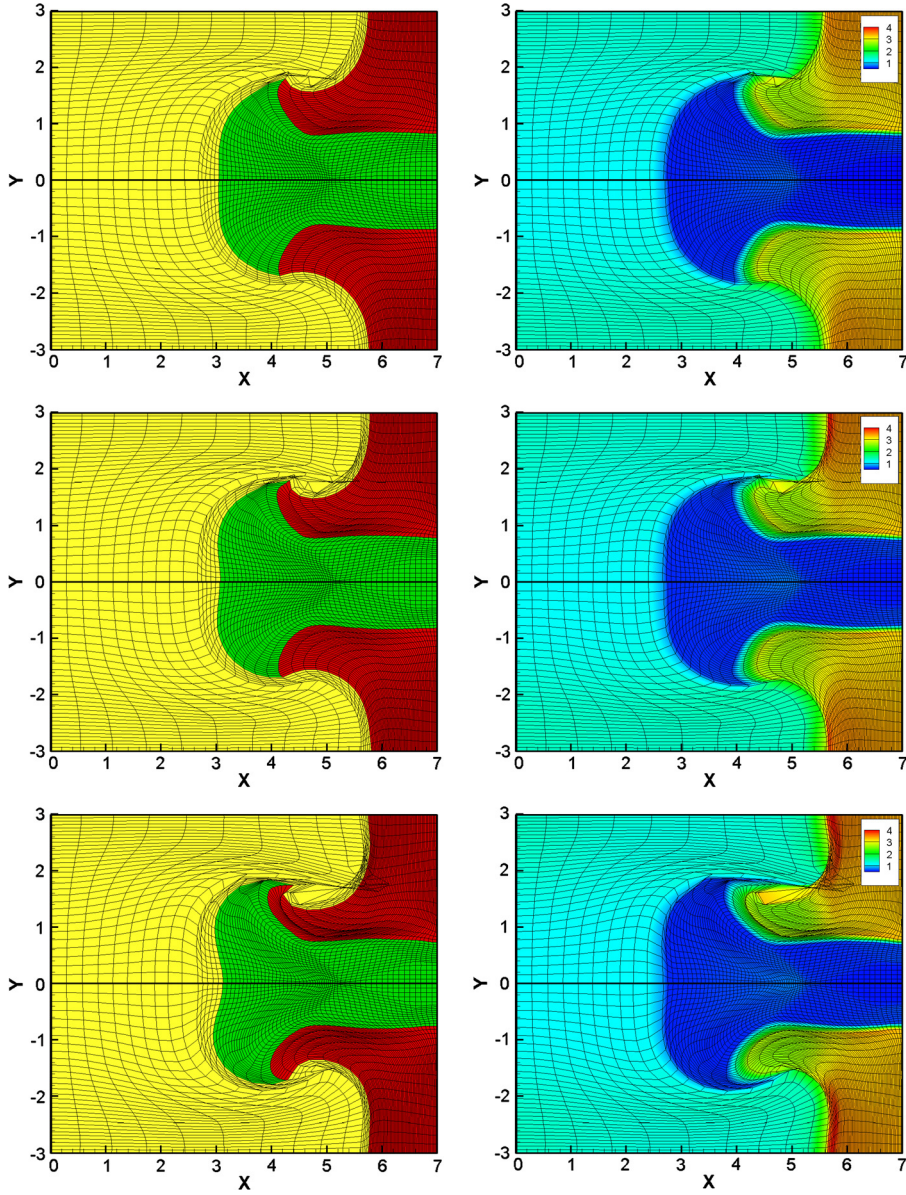


Fig. 17. Triple-point problem at $t = 4.5$. $Y > 0$: scalar viscosity; $Y < 0$: tensor viscosity. Left: colored mesh with different materials; right: internal energy with mesh. Top: first-order scheme; middle: second-order scheme, $\beta = 0.4$; bottom: second-order scheme, $\beta = 0.8$.

distortion. As the dissipation of the tensor viscosity decreases and the vortex rolls up, the mesh only shows a few tangles under the second-order scheme with $\alpha = 0.8$. The results show that the tensor viscosity is applicable for problems that contain physical vorticity. In addition, it is believed that the reconstruction technique could result in spurious mesh motion, and further studies should attempt to dampen this motion for high-order schemes.

6. Concluding remarks

This paper has described a Godunov-type tensor artificial viscosity that is motivated by the scalar form [13]. The proposed method introduces a subcell-based non-negative semidefinite matrix invoking Galilean invariance and entropy inequality. With the construction of the Generalized Riemann Invariant relation in the tensor space, as well as consistency with the GCL, the artificial viscosity was designed in the form of the differential operator μG , where μ is a scalar coefficient. The surface integration of the new artificial viscosity exactly recovers the general form of the subcell-based matrix. In addition, the developed artificial viscosity preserves cylindrical symmetry. Some typical numerical cases have been provided to demonstrate the accuracy, convergence, and robustness of the numerical scheme using our approach.

In the near future, we intend to investigate the extension of the current artificial viscosity and associated Lagrangian schemes to cylindrical coordinates and three dimensions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

This appendix presents a derivation of the discrete velocity gradient operators used in Section 3.3. For a line segment l_{ab} with end points a and b , the unit tangential vector and length of the segment are $\mathbf{e}_{ab}$ and Δl_{ab} . Thus, the projection of the discrete velocity gradient to the segment l_{ab} is

$$G_{ab} = G\mathbf{e}_{ab} = (\mathbf{u}_a - \mathbf{u}_b)/\Delta l_{ab}, \quad (\text{A.1})$$

where G is the velocity gradient and G_{ab} is the projection of G . Using the notation in Section 3.3, the projections of G_1 to segments l_{cc_1} and l_{c_1p} are specified as follows:

$$G_{c_1p} = G_1\mathbf{n}_1 = (\mathbf{u}_{c_1} - \mathbf{u}_p)/\Delta l_{c_1p}, \quad (\text{A.2a})$$

$$G_{cc_1} = G_1\boldsymbol{\tau}_1 = (\mathbf{u}_c - \mathbf{u}_{c_1})/\Delta l_{cc_1}. \quad (\text{A.2b})$$

Define the unit normal vectors as $\mathbf{n}_1 = (\sin\theta, \cos\theta)$ and $\boldsymbol{\tau}_1 = (\cos\theta, -\sin\theta)$. The velocity differences are represented as $\mathbf{u}_{c_1} - \mathbf{u}_p = (\Delta u_{c_1p}, \Delta v_{c_1p})$ and $\mathbf{u}_c - \mathbf{u}_{c_1} = (\Delta u_{cc_1}, \Delta v_{cc_1})$. The matrix form of the above equation is

$$G_1 \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix} = \begin{pmatrix} \Delta u_{c_1p}/\Delta l_{c_1p} & \Delta u_{cc_1}/\Delta l_{cc_1} \\ \Delta v_{c_1p}/\Delta l_{c_1p} & \Delta v_{cc_1}/\Delta l_{cc_1} \end{pmatrix}, \quad (\text{A.3})$$

and then G_1 is computed as

$$\begin{aligned} G_1 &= \begin{pmatrix} \Delta u_{c_1p}/\Delta l_{c_1p} & \Delta u_{cc_1}/\Delta l_{cc_1} \\ \Delta v_{c_1p}/\Delta l_{c_1p} & \Delta v_{cc_1}/\Delta l_{cc_1} \end{pmatrix} \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix} \\ &= \frac{1}{\Delta l_{c_1p}} \Delta \mathbf{u}_{c_1p} \otimes \mathbf{n}_1 + \frac{1}{\Delta l_{cc_1}} \Delta \mathbf{u}_{cc_1} \otimes \boldsymbol{\tau}_1. \end{aligned} \quad (\text{A.4})$$

An expression for G_2 is similarly derived as

$$G_2 = \frac{1}{\Delta l_{c_2p}} \Delta \mathbf{u}_{c_2p} \otimes \mathbf{n}_2 + \frac{1}{\Delta l_{cc_2}} \Delta \mathbf{u}_{cc_2} \otimes \boldsymbol{\tau}_2. \quad (\text{A.5})$$

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